

**RAMCO INSTITUTE OF TECHNOLOGY**  
**Department of Computer Science and Engineering**  
**Academic Year: 2019- 2020 (Odd Semester)**

**Degree, Semester & Branch: V Semester B.E. CSE**

**Course Code & Title: CS8591 Computer Networks**

**Name of the Faculty member: Mrs.M.Swarna Sudha**

**Innovative practice: Concepts Cloning(Collaborative Learning)**

**Topic: Transmission control protocol connection establishment**

**Date and Time: 12.09.2019 & 50 minutes**

**Objective of the activity:**

O1: The students will be able apply the detailed inner workings of TCP.

O2 To distribute the knowledge and experience obtained in concepts to make everybody reach the maximum level of understanding.

O3:To improve listening and communication skills.

**Justification for choosing the particular topic:**

- TCP connection establishment involves many steps and condition which is called as three way handshaking which is difficult to make the students to understand. Teacher used Animated power point explained the establishment and asked the study represent the idea in the form of transition diagram.

**Team formation:**

- A heterogeneous group with nine members was formed. Each group formed by bright students, average students, and slow learners. Classification of students such as a bright student, average students, and slow learners is based on their background, academic performance (bright/average/slow learner) and skill. The team leader will be identified and assigned to each group based on his/her academic performance and skill.
- Guidelines were given to the team leader to represent the ideas in the form of transition diagram
- Finally the state transition diagram was drawn by each group. The activity is conducted on CSE Seminar hall.

**Activity Description**

Concepts cloning is an educational approach to teaching and learning that involves groups of students working together to understand the concepts. It is based on the idea that learning is a naturally social act in which the participants talk among themselves and also share the ideas come out with solution. The faculty gave an idea about the concept that is TCP connection establishment and made the students to draw the diagram for the concepts.



### Reflective report:

#### Challenges:

- Difficulty to make the students understand the concepts
- Tough to create the group
- Always the toppers only represent the concepts

#### Steps taken for address the challenges

- Explain the concept using animated PowerPoint's.
- Make the group of 8 members which contains bright, average and below average.
- Encourage the many students to represent the ideas.

#### Post implementation reflection

- All the students interestingly participated in the activity
- Enhances student satisfaction with the learning experience
- This activity forces students to think more clearly about the concept
- Students were encouraged to correlate the concepts in the form diagram.

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| O1             | 3   | 2   | 2   |     |     |     |     |     |     |      |      | 3    |
| O2             | 2   |     |     |     |     |     |     |     | 3   | 3    |      | 2    |
| O3             | 1   |     |     |     |     |     |     |     | 3   | 3    | 2    | 3    |

Faculty-Incharge

HOD